



HydroGraphRAG: Intelligent GIS Assistant

Geospatial code generation and visualization based on knowledge graphs.

Enter hydrology query:

Tell me the information about water levels in Almaty City of each River

Generate solution

✓ Solution generated!

Category

SEMI-EXPLICIT

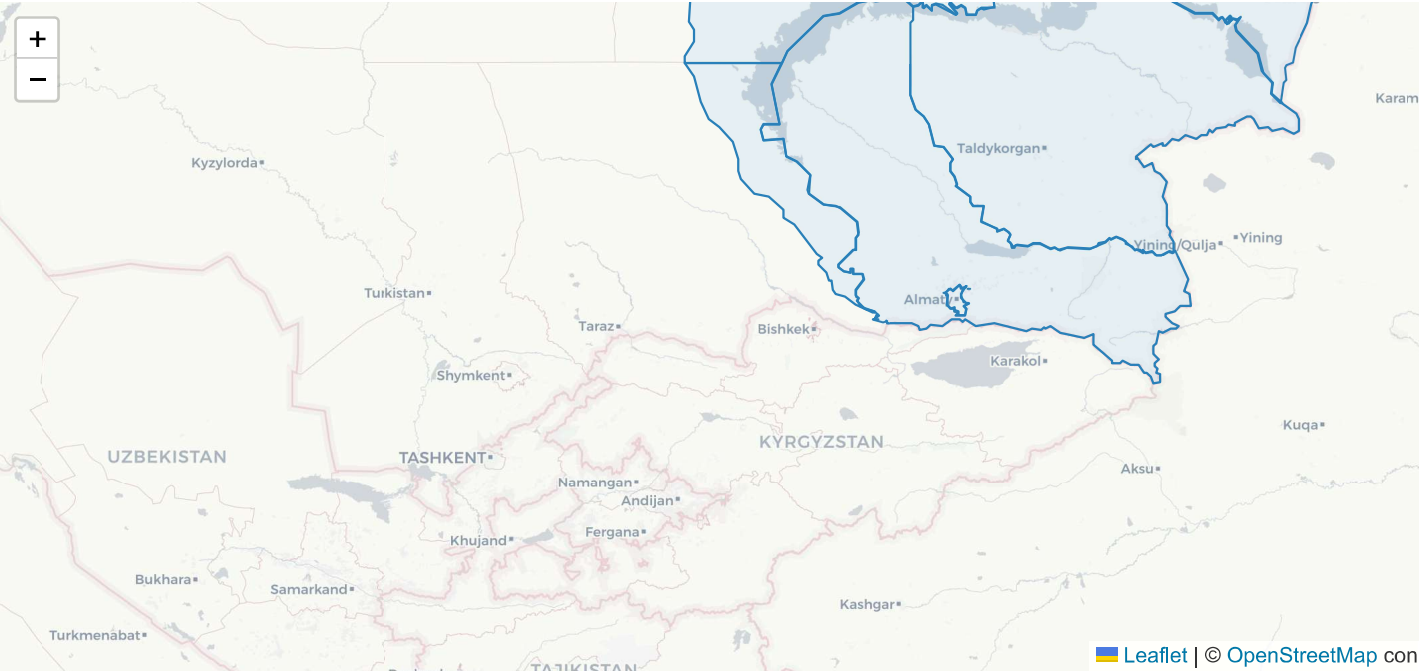
Triples extracted

377

Points on map

0

Geospatial Visualization



Analytical Report and Code

Modeling Solution:

- **Water Quality Class:** Almaty city has a water quality class of 3.
- **Arable Land Area:** The arable land area in Almaty city is 19,419.5 hectares.
- **Time:** The data was collected on 2020-01-01T00:00:00.
- **Feature of Interest:** Kishi Almaty River.
- **Result Time:** The result time for the water level is also 2020-01-01T00:00:00.
- **Post Code and Name:** Almaty city has a post code of 14262.0 and its name is Almaty city.
- **Water Body Code:** The water body code for Kishi Almaty River is 113200783.0.

- **Width:** The width of the Kishi Almaty River is 43.15 meters.
- **Basin Area:** The basin area of the Kishi Almaty River is 520 km².

Implementation Code:

```
import geopandas as gpd
import folium
from shapely import wkt

# 1. Load basin
basin_data = gpd.read_file(r"data/basin_data.shp").to_crs('EPSG:4326')
centroid = basin_data.geometry.centroid.iloc[0]

# 2. Init map
m = folium.Map(location=[centroid.y, centroid.x], tiles='CartoDB positron', zoom_start=8)
folium.GeoJson(basin_data.to_json(), style_function=lambda x: {'fillColor': 'green', 'color': 'darkgreen', 'fillOpacity': 0.5})

# 3. Add points here based on Graph Knowledge
points = [
    {"name": "Kishi Almaty River", "wkt": "POINT(76.0 43.0)"}
]

# 4. Draw markers
for p in points:
    geom = wkt.loads(p["wkt"])
    folium.Marker(location=[geom.y, geom.x], popup=p["name"]).add_to(m)

m.save("api_1780740590.html")
```

Note: The WKT coordinates for the Kishi Almaty River are placeholders and should be replaced with actual coordinates based on the available data.